Iniziato	Friday, 18 June 2021, 13:35
Stato	Completato
Terminato	Friday, 18 June 2021, 14:33
Tempo impiegato	58 min. 39 secondi
Valutazione	Non ancora valutato

Domanda **1**
Completo
Punteggio max.: 6,00

Construct a deterministic TM of the kind you prefer, which decides the following language:

$$\mathcal{L} = \{w \in \{0,1\}^* \mid \text{between any pair of occurrences of 0 in } w$$ there are an odd number of 1s}.

Study the complexity of TM you have defined.

Single-tape TM:

(q_init, ▷) → (q_0, ▷, R)
(q_0, 0) → (q_1, 0, R)
(q_0, 1) → (q_2, 1, R)
(q_0, □) → (q_halt, □, S) // the TM accepts the empty string
(q_1, 1) → (q_3, 1, R)
(q_1, □) → (q_halt, □, S)
(q_2, 0) → (q_1, 0, R)
(q_2, 1) → (q_3, 1, R)
(q_3, 0) → (q_1, 0, R)
(q_3, 1) → (q_2, 1, R)
(q_3, □) → (q_halt, □, S)

Rationale behind q_1, q_2, and q_3:
- q_1: the last symbol found is a 0.
- q_2: the last symbol found is an even occurrence of 1.
- q_3: the last symbol found is an odd occurrence of 1.

The TM above has linear complexity, since it goes through—at most—the whole input string, from beginning to end, one symbol at a time.

Domanda **2**

Completo

Punteggio max.: 7,00

You are required to prove that the following problem $\mathcal{L}$ is in $\mathbf{NP}$. To do that, you can give a TM or define some pseudocode. The language $\mathcal{L}$ includes precisely those binary strings which are encodings of pairs in the form (G, k) where G is a graph and k is a natural number, such that the nodes of G can be assigned a natural number between 1 and k in such a way that nodes which are linked by an edge are assigned distinct natural numbers.

Let G = (V, E) and let l = |⌞(G, k)⌟| be the input encoding length.

• CERTIFICATE: the list N = [n_1,...,n_|V|] of natural numbers assigned to nodes in G.

• CERTIFICATE SIZE: each n_i in N is a natural number <= k, which thus requires O(lgk) bits to be encoded, and there are |V| of them, therefore |⌞N⌟| = O(|V|*lgk), which is polynomial in l.

• CERTIFICATE CHECK: checking that N contains distinct numbers for adjacent nodes requires checking all the edges of G and, for each edge, checking if the nodes involved have distinct numbers in N. Such an algorithm has complexity O(|E|) = O(|V|^2), which is polynomial in l.

From the three points above it follows that the problem is in NP.

Domanda **3**
Completo
Punteggio max.: 7,00

Consider the following problem:

$$\mathtt{1SAT} = \{A \mid A \text{ is a satisfiable 1CNF}\}.$$

To which complexity class does $\mathtt{1SAT}$ belong? Prove your claim.

A 1CNF is a CNF whose clauses contain <= 1 literals, and therefore it is just a conjunction of literals. Such conjunction is satisfiable if and only if all its literals are logically consistent, i.e the formula does not contain two literals which are one the negation of the other.

We can therefore verify the satisfiability of a 1CNF through the following pseudocode:

```
INPUT: L=[l_1,...l_n], the list of literals in the 1CNF
OUTPUT: 1 if the 1CNF is satisfiable, 0 otherwise
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BOOL_VARS ← [ ]
for i ← 1 to n do
   if ⌐L[ i ] in BOOL_VARS then
      return 0
   BOOL_VARS.append(L[ i ])
return 1
```

Remark. We denote by ⌐ the logical negation symbol.

1. INPUT ENCODING
We consider n as the input length, i.e. the number of literals in the 1CNF. A literal, by definition, is either a Boolean variable or its negation. Therefore, in L there are at most as many distinct Boolean variables as there are literals, i.e. L contains O(n) distinct Boolean variables. A first intermediate encoding of L over {0,1, ⌐, #} thus requires O(n) binary encodings of O(n) different symbols—which take O(lgn) bits each—O(n) ⌐s and O(n) #s. The final binary encoding has therefore length l = |⌞L⌟| = O(n*lgn).

2.NUMBER OF INSTRUCTIONS
The number of instructions is dominated by the for loop, which is executed O(n) = O(n*lgn) = O(l) times, polynomial in l.

3.EXECUTION TIME OF EACH INSTRUCTION
The only nontrivial instruction is the append, which takes time O(n) = O(l), polynomial in l.

4. SIZE OF INTERMEDIATE RESULTS
i is an integer number <= n, so |⌞i⌟| = O(lgn) = O(n*lgn) = O(l), polynomial in l.
BOOL_VARS is at most equal to L itself, so |⌞BOOL_VARS⌟| = O(l), polynomial in l.

From the correctness of the pseudocode and the four points above, it follows that 1SAT is in P.

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Vai a...

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